

Chaitanya Dhotre

Cybersecurity Student

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PROFILE SUMMARY

Cybersecurity undergraduate (BSc Hons Cyber Security & Networking, University of Central Lancashire) with a strong foundation in network security, digital forensics, and vulnerability exploitation built through applied coursework and independent projects. Designed and built RogueAP Shield, a Windows endpoint security application implementing a seven-signal rogue access point detection engine with AES-256-GCM encryption and a tamper-evident audit log. Completed structured labs in SQL injection exploitation, SNMP enumeration, OS fingerprinting, packet sniffing, and Windows privilege escalation. Looking for practical experience in real life scenarios to expand my knowledge.

TECHNICAL SKILLS

- **Offensive Security & Pentesting Tools:** Metasploit (msfvenom, exploit/multi/handler), SQLMap, Burp Suite, Nmap (OS detection, service/version scanning), SNMP enumeration (onesixtyone, snmpwalk), ARP spoofing (Ettercap, arpspoof), Wireshark (packet capture & TCP stream analysis)
- **Digital Forensics & Incident Response:** FTK Imager, EnCase Forensic, Autopsy/Sleuth Kit, KAPE, Volatility-class RAM analysis (Magnet RAM Capture, Belkasoft Live RAM Capture), Regshot, Process Monitor/Process Hacker, WinHex, registry & NTFS artefact analysis (MFT, \$UsnJrnl, Prefetch, Shellbags)
- **Malware Analysis:** Static analysis (PE header inspection, entropy analysis, string extraction), dynamic sandbox analysis, Ghidra, PEiD/pestudio, steganography detection (StegSpy, OpenStego, binwalk, ExifTool)
- **Networking & Systems:** TCP/IP, subnetting, LAN architecture, firewall & IDS/IPS concepts, Windows Server & Active Directory fundamentals, Linux (CachyOS/Arch/Fedora), Windows Filtering Platform (WFP), Wi-Fi security (WPA2/WPA3, 802.11 beacon frame analysis)
- **Programming & Scripting:** C#, C++, Java (foundational), Python (in progress), HTML/CSS, SQL, Git/GitHub
- **Cryptography & Secure Design:** AES-256-GCM, PBKDF2-SHA256, SHA-256 hash chaining, CIA triad application, secure key management (DEK/KEK architecture)

PROJECTS

Cybercrime Digital Forensics Investigation — Marriott Starwood Breach Case Study | *NCC Education Coursework*

- Produced a full digital forensics investigation plan for a simulated 500-million-record data breach, covering cybercrime classification under the Budapest Convention, CFAA, and UK Computer Misuse Act.
- Mapped evidence acquisition strategy across employee workstations, an Exchange mail server, and a MySQL Linux server using FTK Imager, EnCase Forensic, dd/dc3dd, and KAPE, following RFC 3227 order-of-volatility principles.
- Detailed detection of a steganography-based anti-forensic technique (LSB steganography hiding a keylogger inside an email image attachment) and the corresponding detection workflow using StegSpy, OpenStego, and binwalk.

RogueAP Shield — Rogue Access Point Detection Framework

- Designed and built a Windows endpoint security application in .NET 9 that detects "evil twin" Wi-Fi attacks in real time using a weighted, seven-signal risk-scoring engine (BSSID baseline matching, 802.11 beacon fingerprinting, gateway TTL anomaly detection, DNS resolution probing, and signal-strength outlier analysis).
- Implemented AES-256-GCM encrypted local storage, a SHA-256 hash-chained tamper-evident audit log, and an automatic WireGuard VPN kill switch enforced at the kernel level via the Windows Filtering Platform.

- Built a no-configuration auto-learning mode to baseline trusted networks in home, enterprise, and campus environments, removing the need for manual BSSID whitelisting.
- Validated the system through controlled attack simulations, including a live evil-twin attack recreated with a mobile hotspot, and multi-AP enterprise testing across 53 access points on a university campus network; all 15 functional and 7 non-functional requirements were met.

Vulnerability Exploitation & Network Penetration Testing Labs

- Documented a full exploitation chain for CVE-2019-20183 (unauthenticated SQL injection), from Nmap/Burp Suite reconnaissance through automated exploitation with SQLMap, including database enumeration and privilege escalation.
- Planned SNMP enumeration and community-string brute-forcing methodology using onesixtyone and snmpwalk, and produced hardening countermeasures (SNMPv3 migration, ACLs, port restriction).
- Detailed an ARP-spoofing packet-sniffing attack against an FTP server using Ettercap and Wireshark to demonstrate cleartext credential exposure, paired with defensive recommendations (SFTP/FTPS migration, Dynamic ARP Inspection).
- Mapped a Windows unquoted service path privilege escalation exploit chain using msfvenom and a Metasploit reverse-shell listener to achieve SYSTEM-level access, with full remediation guidance.

Kuch Bhi — End-to-End Encrypted Messaging Application | *(In Development)*

- Architecting a cross-platform encrypted messaging app in Flutter with a feature-first clean architecture (presentation → repository → data → domain layers) and offline-first local persistence via Drift/SQLite.
- Implementing end-to-end message encryption (AES-256-GCM with X25519 key exchange) with all cryptographic operations isolated to background threads to keep the UI responsive.
- Built a real-time backend on a self-hosted PocketBase server with Firebase Cloud Messaging for push notifications, Google OAuth authentication, and a delivery/read-receipt state machine (sent/delivered/read).
- Currently implementing media handling (compressed photo/video sharing, voice messages) with server-side expiry policies ahead of group-chat encryption and adaptive tablet/foldable UI phases.

EDUCATION

Level 5 Diploma in Computing (Cyber Security)

NCC Education (UK)

Completion: 2026

BSc (Hons) Cyber Security and Networking

University of Central Lancashire (via VUE)

Completion: 2027